

WEAPON DETECTION FROM BODYCAM

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Generative AI and Computer Vision, Semester A

MOTIVATION

Detect weapons from body-worn cameras for military and public safety applications.

- enhanced situational awareness and efficient post event analysis.
- For some images, we applied bodycam distortions and a 'realism pass'.

WEAPON DETECTION: NOVELTY

In surveillance footage and general images

- Bodycam images dataset (table)

In first-person video: new challenges

- Strong motion blur from rapid body movement.
- Extreme viewpoint changes and unusual angles
- Very short and partial visibility of weapons
- Highly dynamic, cluttered scenes
- Rapidly changing lighting conditions

No labeled training/evaluation dataset exists for body-cam weapon detection

- Create synthetically using Generative models

Dataset	Modality
Weapon Detection CCTV v3 – CCTV-oriented images (weapon classes)	Images / CCTV-style
Weapon Detection (Kaggle) – multi-class weapons	Images
Weapon Detection (alternate Kaggle) – general weapon images	Images
Pistols Object Detection (Roboflow)	Images
Weapons in Images Dataset	Images + some CCTV frames



DATASETS REQUIREMENTS

Labeled realistic body-cam images

- An image and a bounding box (ground truth label) for object detection

Alternatives for Synthetic data generation

- Augment existing dataset: introduce distortions to existing surveillance images with already labeled weapons.
- Inpainting: use existing FPV footage and in-paint weapons into target locations
- Generating from scratch/distillation: generate or augment body-cam images with a weapon, use a strong model (SAM) to label/detect weapons

ML TASK DEFINITION

Input

An image: a single frame from a body cam with appropriate resolution

Include realistic distortions: blur, partial occlusion, imperfect light, cluttered scene

Output

Input image with labeled boxes where there is weapons.

KPIs/Metrics

Object detection metrics: mAP@50

Classification metrics: accuracy

Mission-critical task: increase the importance of false negatives

MODELS

Baseline

Pretrained on surveillance or general images (Open Images (V6 / V7)

Lower bound of model performance

Fine-tuned models

Fine-tune using synthetic data

Using YOLOv8s (Small) model

RT-DETR

Metric	Value
mAP ₅₀	XX
Precision	XX
Recall	XX
Inference Speed	XX

As a gold standard

The best possible case, upper bound of model performance

SAM or Grounding Dino

Heavy model that can't run in real-time over a mobile camera

THANK YOU